Solved selected problems of Introductory functional analysis with applications - Erwin Kreyszig

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Chapter 3 - Inner Product Spaces. Hilbert Spaces

3.1 - Inner Product Space. Hilbert Space

Proof. **1** Let us prove (4) as follows

$$\begin{aligned}
\|x+y\|^2 + \|x-y\|^2 &= \langle x+y, x+y\rangle + \langle x-y, x-y\rangle \\
&= \langle x,x\rangle + \langle x,y\rangle + \langle y,x\rangle + \langle y,y\rangle + \langle x,x\rangle - \langle x,y\rangle - \langle y,x\rangle + \langle y,y\rangle \\
&= 2\langle x,x\rangle + 2\langle y,y\rangle \\
&= 2(\|x\|^2 + \|y\|^2)
\end{aligned}$$

Where we used the definition of the norm $\|x\| = \sqrt{\langle x,x\rangle}$ and the distributive property of the inner products. $\square$

Proof. **2** Let $x \perp y$ in an inner product space X then

$$\begin{aligned}
\|x+y\|^2 &= \langle x+y, x+y\rangle \\
&= \langle x,x\rangle + \langle x,y\rangle + \langle y,x\rangle + \langle y,y\rangle \\
&= \langle x,x\rangle + \langle y,y\rangle \\
&= \|x\|^2 + \|y\|^2
\end{aligned}$$

Where we used that $\langle x,y\rangle = \langle y,x\rangle = 0$ since $x \perp y$.

Suppose now we have m mutually orthogonal vectors, i.e. $x_1 \perp x_2 \perp ... \perp x_m$ then

$$\begin{aligned}
\|x_1 + x_2 + ... + x_m\|^2 &= \langle x_1 + x_2 + ... + x_m, x_1 + x_2 + ... + x_m\rangle \\
&= \langle x_1, x_1\rangle + \langle x_1, x_2\rangle + ... + \langle x_m, x_{m-1}\rangle + \langle x_m, x_m\rangle \\
&= \langle x_1, x_1\rangle + \langle x_2, x_2\rangle + ... + \langle x_m, x_m\rangle \\
&= \|x_1\|^2 + \|x_2\|^2 + ... + \|x_m\|^2
\end{aligned}$$

Where we used that $\langle x_i, x_j\rangle = 0$ for $i \neq j$ since they are orthogonal. $\square$

Proof. **3** Let $x, y \in \mathbb{R}$ and suppose we have the following property

$$\|x+y\|^2 = \|x\|^2 + \|y\|^2$$

then since $\|x\|^2 = \langle x, x\rangle$ above equation implies that

$$\begin{aligned} \langle x+y, x+y\rangle &= \langle x, x\rangle + \langle y, y\rangle \\ \langle x, x\rangle + \langle x, y\rangle + \langle y, x\rangle + \langle y, y\rangle &= \langle x, x\rangle + \langle y, y\rangle \\ \langle x, y\rangle + \langle y, x\rangle &= 0 \end{aligned}$$

But since $x, y \in \mathbb{R}$ we have that $\langle x, y\rangle = \langle y, x\rangle$, hence

$$\langle x, y\rangle = \langle y, x\rangle = 0$$

Therefore $x \perp y$.

Let now $x, y \in \mathbb{C}$ and suppose that the property holds in this case too. Then as before must be that

$$\langle x, y\rangle + \langle y, x\rangle = 0$$

Also, we know that $\langle y, x\rangle = \overline{\langle x, y\rangle}$ hence we need that

$$\langle x, y\rangle = -\overline{\langle x, y\rangle}$$

But since we are assuming a complex inner product we have that $\langle x, y\rangle$ is of the form $a + bi$ hence

$$a + bi = -(a - bi) = -a + bi$$

Equating the real and imaginary parts we see that $a = 0$ and $b = b$. So for example if $x = 1 + i$ and $y = 1 - i$ we see that the Pythagorean property holds, as we show below

$$\|1 + i + 1 - i\|^2 = \|2\|^2 = 4 = 2 + 2 = \|1 + i\|^2 + \|1 - i\|^2$$

But inner product gives us

$$\langle x, y\rangle = (1 + i)\overline{(1 - i)} = (1 + i)(1 + i) = 1 + i + i - 1 = 2i$$

Therefore x and y are not orthogonal. $\square$

Proof. **6** Let $x \neq 0$ and $y \neq 0$

(a) Let $x \perp y$ we want to prove that $\{x, y\}$ is a linearly independent set i.e. we want to prove

$$\alpha_1 x + \alpha_2 y = 0$$

Only for $\alpha_1 = \alpha_2 = 0$. Suppose the opposite we want to arrive at a contradiction, let $\alpha_1 \neq 0$ and $\alpha_2 = 0$ then $\alpha_1 x + 0y = 0$ i.e. $x = 0$ but we said that $x \neq 0$ so it must be that $\alpha_1 \neq 0$ and $\alpha_2 \neq 0$. Then we can write that

$$x = -\frac{\alpha_2}{\alpha_1} y$$

Also, we know that $\langle x, y \rangle = 0$ since $x \perp y$ then

$$\langle x, y \rangle = \left\langle -\frac{\alpha_2}{\alpha_1} y, y \right\rangle = -\frac{\alpha_2}{\alpha_1} \langle y, y \rangle = 0$$

But this implies that $y = 0$ and we said that $y \neq 0$, a contradiction.

Therefore must be that $\{x, y\}$ is a linearly independent set.

(b) Now, let $\{x_1, ..., x_m\}$ be a set of mutually orthogonal non-zero vectors, we want to show they are a linearly independent set.

As before, suppose the opposite, we want to arrive at a contradiction.

If $\alpha_i \neq 0$ for only the ith vector then $x_i = 0$ and that cannot happen.

So let us assume that $\alpha_i \neq 0$ for $1 \leq i \leq m$, if not all α_is were different from zero the following still applies, then

$$\begin{aligned} \alpha_1 x_1 + ... + \alpha_n x_n = 0 \\ x_1 = -\frac{\alpha_2}{\alpha_1} x_2 - ... - \frac{\alpha_n}{\alpha_1} x_n \end{aligned}$$

Also we know that $\langle x_1, x_2 \rangle = 0$ then

$$\begin{aligned} \langle x_1, x_2 \rangle &= \left\langle -\frac{\alpha_2}{\alpha_1} x_2 - ... - \frac{\alpha_n}{\alpha_1} x_n, x_2 \right\rangle \\ &= -\frac{\alpha_2}{\alpha_1} \langle x_2, x_2 \rangle - ... - \frac{\alpha_n}{\alpha_1} \langle x_n, x_2 \rangle \\ &= -\frac{\alpha_2}{\alpha_1} \langle x_2, x_2 \rangle \\ &= 0 \end{aligned}$$

Where we used that $\langle x_i, x_2 \rangle = 0$ for $1 \leq i \leq m$ and $i \neq 2$. But the last equality implies that $x_2 = 0$ which is a contradiction.

Therefore must be that the orthogonal nonzero vectors $\{x_1, ..., x_m\}$ are a linearly independent set.

$\square$

Proof. **7** Let $\langle x, u\rangle = \langle x, v\rangle$ for all x, then must also be true for $x = u$ i.e. $\langle u, u\rangle = \langle u, v\rangle$ and for $x = v$ i.e. $\langle v, u\rangle = \langle v, v\rangle$.
Now, let us compute $\langle u - v, u - v\rangle$ as follows

$$\begin{aligned}\langle u - v, u - v\rangle &= \langle u, u\rangle - \langle u, v\rangle - \langle v, u\rangle + \langle v, v\rangle \\ &= \langle u, v\rangle - \langle u, v\rangle - \langle v, u\rangle + \langle v, u\rangle \\ &= 0\end{aligned}$$

But then this implies that $u - v = 0$. Therefore $u = v$. $\square$

Proof. **8** Let us compute (9) knowing that $\|x\| = \sqrt{\langle x, x\rangle}$ then

$$\begin{aligned}\frac{1}{4}(\|x+y\|^2 - \|x-y\|^2) &= \frac{1}{4}(\langle x+y, x+y\rangle - \langle x-y, x-y\rangle) \\ &= \frac{1}{4}(\langle x, x\rangle + \langle x, y\rangle + \langle y, x\rangle + \langle y, y\rangle - \\ &\quad - (\langle x, x\rangle + \langle x, -y\rangle + \langle -y, x\rangle + \langle -y, -y\rangle)) \\ &= \frac{1}{4}(\langle x, x\rangle + \langle x, y\rangle + \langle y, x\rangle + \langle y, y\rangle - \\ &\quad - \langle x, x\rangle + \langle x, y\rangle + \langle y, x\rangle - \langle y, y\rangle) \\ &= \frac{1}{4}(2\langle x, y\rangle + 2\langle x, y\rangle) \\ &= \langle x, y\rangle\end{aligned}$$

Where we used that $\langle x, y\rangle = \langle y, x\rangle$ since we are in a real inner product space. $\square$

Proof. **9** Let us compute (10) knowing that we are in a complex inner product space, then

$$\begin{aligned}
\frac{1}{4}(\|x+y\|^2-\|x-y\|^2) &= \frac{1}{4}(\langle x+y,x+y\rangle-\langle x-y,x-y\rangle)\\
&= \frac{1}{4}(2\langle x,y\rangle+2\langle y,x\rangle)\\
&= \frac{1}{4}(2\langle x,y\rangle+2\overline{\langle x,y\rangle})\\
&= \frac{1}{2}(\mathrm{Re}\langle x,y\rangle+i\mathrm{Im}\langle x,y\rangle+\mathrm{Re}\langle x,y\rangle-i\mathrm{Im}\langle x,y\rangle)\\
&= \mathrm{Re}\langle x,y\rangle
\end{aligned}$$

Now, we compute the second equation as follows

$$\begin{aligned}
\frac{1}{4}(\|x+iy\|^2-\|x-iy\|^2) &= \frac{1}{4}(\langle x+iy,x+iy\rangle-\langle x-iy,x-iy\rangle)\\
&= \frac{1}{4}(\langle x,x\rangle+\langle x,iy\rangle+\langle iy,x\rangle+\langle iy,iy\rangle-\\
&\quad-(\langle x,x\rangle+\langle x,-iy\rangle+\langle -iy,x\rangle+\langle -iy,-iy\rangle))\\
&= \frac{1}{4}(\langle x,x\rangle-i\langle x,y\rangle+i\langle y,x\rangle+\langle y,y\rangle-\\
&\quad-\langle x,x\rangle-i\langle x,y\rangle+i\langle y,x\rangle-\langle y,y\rangle)\\
&= \frac{1}{2}(-i\langle x,y\rangle+i\langle y,x\rangle)\\
&= \frac{1}{2}(-i\langle x,y\rangle+i\overline{\langle x,y\rangle})\\
&= \frac{1}{2}(-i\mathrm{Re}\langle x,y\rangle+\mathrm{Im}\langle x,y\rangle+i\mathrm{Re}\langle x,y\rangle+\mathrm{Im}\langle x,y\rangle)\\
&= \mathrm{Im}\langle x,y\rangle
\end{aligned}$$

□

Proof. **10** Let z_1 and z_2 be complex numbers and let $\langle z_1, z_2 \rangle = z_1\overline{z_2}$ we want to prove that in fact this is a valid definition of an inner product.
So if this expression defines an inner product, it must have the four properties of inner products. Then for (IP1) let z_3 be also a complex number

$$\langle z_1 + z_3, z_2 \rangle = (z_1 + z_3)\overline{z_2} = z_1\overline{z_2} + z_3\overline{z_2} = \langle z_1, z_2 \rangle + \langle z_3, z_2 \rangle$$

For (IP2) we see that

$$\langle \alpha z_1, z_2 \rangle = \alpha z_1\overline{z_2} = \alpha \langle z_1, z_2 \rangle$$

For (IP3) we see that

$$\langle z_1, z_2 \rangle = z_1\overline{z_2} = \overline{\overline{z_1}z_2} = \overline{\langle z_2, z_1 \rangle}$$

Finally for (IP4) we see that $\langle z_1, z_1 \rangle = z_1\overline{z_1} \geq 0$. Also, let $\langle z_1, z_1 \rangle = 0$ then

$$z_1\overline{z_1} = (x + iy)(x - iy) = x^2 + y^2 = 0$$

Then the only way for this equation to hold is that $x = y = 0$ and hence $z_1 = 0$. Now, if we suppose $z_1 = 0$ then $\langle z_1, z_1 \rangle = z_1\overline{z_1} = 0$ and hence (IP4) is also satisfied.
Therefore $\langle z_1, z_1 \rangle = z_1\overline{z_1}$ defines in fact an inner product.
Also, to have orthogonality it must happen that $\langle z_1, z_2 \rangle = z_1\overline{z_2} = 0$. $\square$

Proof. **11** Let X be the vector space of all ordered pairs of complex numbers. The norm defined as $\|x\| = |\xi_1| + |\xi_2|$ cannot be obtained from an inner product. We prove this by showing that the norm does not satisfy the parallelogram equality. Let us take $x = (-1, 1)$ and $y = (-1, -1)$ then

$$2(\|x\|^2 + \|y\|^2) = 2((|-1| + |1|)^2 + (|-1| + |-1|)^2) = 16$$

But

$$\|x + y\|^2 + \|x - y\|^2 = (|-1-1| + |1-1|)^2 + (|-1+1| + |1+1|)^2 = 8$$

Therefore the parallelogram equality is not satisfied and hence the norm defined cannot be obtained from an inner product. $\square$

Proof. **13** The inner product defined in 3.1-5 for continuous functions states that

$$\langle x, y \rangle = \int_a^b x(t)\overline{y(t)}\ dt$$

Let us check that this definition satisfies (IP1) to (IP4). We assume $x(t)$ and $y(t)$ complex-valued functions.
(IP1):

$$\begin{aligned} \langle x + y, z \rangle &= \int_a^b (x(t) + y(t))\overline{z(t)}\ dt \\ &= \int_a^b x(t)\overline{z(t)}\ dt + \int_a^b y(t)\overline{z(t)}\ dt \\ &= \langle x, z \rangle + \langle y, z \rangle \end{aligned}$$

(IP2):

$$\langle \alpha x, y \rangle = \int_a^b \alpha x(t)\overline{y(t)}\ dt = \alpha \int_a^b x(t)\overline{y(t)}\ dt = \alpha \langle x, y \rangle$$

(IP3):

$$\langle x, y \rangle = \int_a^b x(t)\overline{y(t)}\ dt = \int_a^b \overline{\overline{x(t)}y(t)}\ dt = \overline{\langle y, x \rangle}$$

(IP4):

$$\langle x, x \rangle = \int_a^b x(t)\overline{x(t)}\ dt = \int_a^b |x(t)|^2\ dt \geq 0$$

Let now $\langle x, x \rangle = 0$ then

$$\int_a^b |x(t)|^2\ dt = 0$$

Then must be that $x(t) = 0$ since $x(t)$ is continuous. Conversely, if we suppose that $x(t) = 0$ we get that

$$\langle x, x \rangle = \int_a^b 0 \cdot 0\ dt = 0$$

Therefore the inner product defined in 3.1-5 satisfy the properties (IP1) to (IP4). □

3.2 - Further Properties of Inner Produet Spaces

Proof. **2** An example of subspace of l^2 is the set sequences with finitely many entries different from zero followed by zeroes i.e.

$$Y = \{(x_1, x_2, ..., x_k, 0, 0, 0, ...) : x_i \in \mathbb{R}, 0 \leq i \leq k\}$$

Suppose $x = (x_1, x_2, ..., x_k, 0, 0, 0, ...) \in Y$ then

$$\|x\|_2 = \left(\sum_{i=1}^{\infty} |x_i|^2\right)^{1/2} = \left(\sum_{i=1}^{k} |x_i|^2\right)^{1/2} < \infty$$

So $Y \subset l^2$.
On the other hand, let $x' = (x'_1, x'_2, ..., x'_k, 0, 0, 0, ...) \in Y$, and α, β be scalars then

$$\begin{aligned} \alpha x + \beta x' &= \alpha(x_1, x_2, ..., x_k, 0, 0, 0, ...) + \beta(x'_1, x'_2, ..., x'_k, 0, 0, 0, ...) \\ &= (\alpha x_1, \alpha x_2, ..., \alpha x_k, 0, 0, 0, ...) + (\beta x'_1, \beta x'_2, ..., \beta x'_k, 0, 0, 0, ...) \\ &= (\alpha x_1 + \beta x'_1, \alpha x_2 + \beta x'_2, ..., \alpha x_k + \beta x'_k, 0, 0, 0, ...) \end{aligned}$$

We see that $\alpha x_i + \beta x'_i \in \mathbb{R}$ since $\mathbb{R}$ is a vector space then Y is a subspace of l^2. $\square$

Proof. **3** Let X be the inner product space consisting of the polynomial $x = 0$ and all real polynomials in t of degree not exceeding 2, considered for real $t \in [a, b]$ with inner product defined as

$$\langle x, y\rangle = \int_a^b x(t)y(t)\ dt$$

Let $(x_n) = (\alpha_n t^2 + \beta_n t + \gamma_n) \in X$ be a Cauchy sequence on X where $\alpha_n, \beta_n, \gamma_n \in \mathbb{R}$ then since this is a Cauchy sequence, given $\epsilon > 0$ we can find $N \in \mathbb{N}$ such that when $n, m > N$ we have that

$$\|x_n - x_m\| = \left(\int_a^b (x_n(t) - x_m(t))^2\ dt\right)^{1/2} < \epsilon$$

Hence

$$\left(\int_a^b ((\alpha_n - \alpha_m)t^2 + (\beta_n - \beta_m)t + (\gamma_n - \gamma_m))^2\ dt\right)^{1/2} < \epsilon$$

For this integral to tend to 0 over the fixed interval $[a, b]$ since t does not depend on n must happen that $(\alpha_n - \alpha_m) \to 0$, $(\beta_n - \beta_m) \to 0$ and $(\gamma_n - \gamma_m) \to 0$.
This implies that (α_n), (β_n) and (γ_n) are Cauchy sequences of real numbers which we know they converge, let us say that they converge to α, β and γ respectively.
Now, let us define $x = \alpha t^2 + \beta t + \gamma$ then

$$\begin{aligned}\|x_n - x\| &= \left(\int_a^b (x_n(t) - x(t))^2\ dt\right)^{1/2} \\ &= \left(\int_a^b ((\alpha_n - \alpha)t^2 + (\beta_n - \beta)t + (\gamma_n - \gamma))^2\ dt\right)^{1/2}\end{aligned}$$

But we know that $\alpha_n \to \alpha$, $\beta_n \to \beta$ and $\gamma_n \to \gamma$ therefore we have that

$$\left(\int_a^b ((\alpha_n - \alpha)t^2 + (\beta_n - \beta)t + (\gamma_n - \gamma))^2\ dt\right)^{1/2} \to 0$$

Or what is the same $x_n \to x$ and hence since x_n was a Cauchy sequence then X is complete.

Let now Y be all $x \in X$ such that $x(a) = 0$. Let $y, y' \in Y$ and let α, β be scalars. We know that when $t = a$ then $y(a) = y'(a) = 0$ then for $z(t) = \alpha y(t) + \beta y'(t)$ we have that $z(a) = \alpha y(a) + \beta y'(a) = 0 + 0 = 0$. Therefore Y is a subspace of X.

Finally, let Z be all $x \in X$ of degree 2. Let $z, z' \in Z$ and let λ, ρ be scalars. Then let us compute $\lambda z + \rho z'$ as follows

$$\lambda(\alpha t^2 + \beta t + \gamma) + \rho(\alpha' t^2 + \beta' t + \gamma') = (\lambda\alpha + \rho\alpha')t^2 + (\lambda\beta + \rho\beta')t + (\lambda\gamma + \rho\gamma')$$

Where $\alpha, \alpha', \beta, \beta', \gamma$ and γ' are not zero. Then we see that if $\lambda = \rho = 0$ we get that $\lambda z + \rho z' = 0$ but the polynomial $z = 0$ is not included in Z then Z is not a subspace of X. $\square$

Proof. **4** Let $y \perp x_n$ and $x_n \to x$ then by the Continuity of the Inner Product we have that

$$\langle x_n, y \rangle \to \langle x, y \rangle$$

This implies that

$$\| \langle x_n, y \rangle - \langle x, y \rangle \| \to 0$$

But we know that $\langle x_n, y \rangle = 0$ because $y \perp x_n$ hence

$$\| \langle x, y \rangle \| \to 0$$

And by the (N2) property for norms, we get that

$$\langle x, y \rangle = 0$$

Therefore $x \perp y$. $\square$

Proof. **5** Let a sequence (x_n) in an inner product space such that $\|x_n\| \to \|x\|$ and $\langle x_n, x\rangle \to \langle x, x\rangle$. Given that $\|x_n\| \to \|x\|$ then $\|x_n\|^2 \to \|x\|^2$ so given $\epsilon^2/3 > 0$ there is $N \in \mathbb{N}$ such that when $n \geq N$ we see that

$$|\|x_n\|^2 - \|x\|^2| = |\langle x_n, x_n\rangle - \langle x, x\rangle| < \epsilon^2/3$$

On the other hand, since $\langle x_n, x\rangle \to \langle x, x\rangle$ given $\epsilon^2/3 > 0$ there is $N' \in \mathbb{N}$ such that when $n \geq N'$ we get that

$$|\langle x_n, x\rangle - \langle x, x\rangle| = |\langle x_n - x, x\rangle| < \epsilon^2/3$$

Also, let us note that

$$\begin{aligned}
\|x_n - x\|^2 &= |\langle x_n - x, x_n - x\rangle| \\
&= |\langle x_n - x, x_n\rangle - \langle x_n - x, x\rangle| \\
&\leq |\langle x_n - x, x_n\rangle| + |\langle x_n - x, x\rangle| \\
&\leq |\langle x_n, x_n\rangle - \langle x, x_n\rangle| + |\langle x_n - x, x\rangle| \\
&\leq |\langle x_n, x_n\rangle - \langle x, x\rangle + \langle x, x\rangle - \langle x, x_n\rangle| + |\langle x_n - x, x\rangle| \\
&\leq |\langle x_n, x_n\rangle - \langle x, x\rangle| + |\langle x, x - x_n\rangle| + |\langle x_n - x, x\rangle|
\end{aligned}$$

Then if we take $M = \max(N, N')$ when $n \geq M$ we have that

$$\begin{aligned}
\|x_n - x\|^2 &\leq |\langle x_n, x_n\rangle - \langle x, x\rangle| + |\langle x, x - x_n\rangle| + |\langle x_n - x, x\rangle| \\
&< \epsilon^2/3 + \epsilon^2/3 + \epsilon^2/3 = \epsilon^2
\end{aligned}$$

Therefore $\|x_n - x\| < \epsilon$ and hence $x_n \to x$. □

Proof. **9** Let V be the vector space of all continuous complex-valued functions on $J = [a, b]$. Let $X_1 = (V, \|\cdot\|_\infty)$ where $\|x\|_\infty = \max_{t\in J} |x(t)|$ and let $X_2 = (V, \|\cdot\|_2)$ where

$$\|x\|_2 = \sqrt{\langle x, x\rangle} \qquad \langle x, y\rangle = \int_a^b x(t)\overline{y(t)}\ dt$$

Also, let us define $i : X_1 \to X_2$ be the identity mapping $x \to x$.
Suppose we take $x, y \in V$ then we see that

$$\begin{aligned}\|x - y\|_2 &= \sqrt{\langle x - y, x - y\rangle}\\ &= \sqrt{\int_a^b (x(t) - y(t))(\overline{x(t) - y(t)})\ dt}\\ &= \sqrt{\int_a^b |x(t) - y(t)|^2\ dt}\end{aligned}$$

But also we have that

$$\sqrt{\int_a^b |x(t) - y(t)|^2\ dt} \leq \sqrt{\int_a^b \max_{t\in J} |x(t) - y(t)|^2\ dt} \ = \max_{t\in J} |x(t) - y(t)|\sqrt{b - a}$$

So given $\epsilon > 0$ if we take $\delta = \epsilon/\sqrt{b - a}$ such that

$$\|x - y\|_\infty = \max_{t\in J} |x(t) - y(t)| < \delta = \frac{\epsilon}{\sqrt{b - a}}$$

We get that

$$\|x - y\|_2 \leq \max_{t\in J} |x(t) - y(t)|\sqrt{b - a} < \epsilon$$

Therefore the identity mapping i is continuous.

$\square$

Proof. **10** Let $T : X \to X$ be a bounded linear operator on a complex inner product space X and let $\langle Tx, x \rangle = 0$ for all $x \in X$.
Let $x = \alpha u + v \in X$ for some $u, v \in X$ and $\alpha \in \mathbb{C}$ then

$$\begin{aligned}
\langle Tx, x \rangle &= \langle \alpha Tu + Tv, \alpha u + v \rangle \\
&= \langle \alpha Tu, \alpha u + v \rangle + \langle Tv, \alpha u + v \rangle \\
&= \alpha\overline{\alpha} \langle Tu, u \rangle + \alpha \langle Tu, v \rangle + \overline{\alpha} \langle Tv, u \rangle + \langle Tv, v \rangle \\
&= \alpha \langle Tu, v \rangle + \overline{\alpha} \langle Tv, u \rangle
\end{aligned}$$

Then taking $\alpha = 1$ we get that

$$\begin{aligned}
\langle Tu, v \rangle + \langle Tv, u \rangle &= 0 \\
\langle Tu, v \rangle &= - \langle Tv, u \rangle
\end{aligned}$$

But if we take $\alpha = i$ we have that $i \langle Tu, v \rangle - i \langle Tv, u \rangle = 0$ or

$$\begin{aligned}
\langle Tu, v \rangle - \langle Tv, u \rangle &= 0 \\
\langle Tu, v \rangle &= \langle Tv, u \rangle
\end{aligned}$$

This implies that $\langle Tu, v \rangle = 0$ for all $u, v \in X$ so in particular if we select $v = Tu$ then $\langle Tu, Tu \rangle = 0$ and hence by (IP4) we have that $Tu = 0$. Therefore $T = 0$.

Let us consider now the Euclidean plane, a real inner product space, and let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by

$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \cos \pi/2 & -\sin \pi/2 \\ \sin \pi/2 & \cos \pi/2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -y \\ x \end{bmatrix}$$

Then let us compute the following inner product

$$\left\langle T \begin{bmatrix} x \\ y \end{bmatrix}, \begin{bmatrix} x \\ y \end{bmatrix} \right\rangle = \left\langle \begin{bmatrix} -y \\ x \end{bmatrix}, \begin{bmatrix} x \\ y \end{bmatrix} \right\rangle = -yx + xy = 0$$

Therefore, we see that the inner product is 0 but $T \neq 0$ so above result does not hold for real inner product spaces. $\square$

3.3 - Orthogonal Complements and Direct Sums

Proof. **1** Let H be a Hilbert space, $M \subset H$ a convex subset, and (x_n) a sequence in M such that $\|x_n\| \to d$ where $d = \inf_{x\in M} \|x\|$.
Let us consider $\overline{M}$ the closure of M then $\overline{M}$ is complete because $\overline{M} \subset H$, also, $(x_n) \in \overline{M}$.
Let $n, m \in \mathbb{N}$ where $n \geq m$. We see that $\frac{1}{2}x_n + \frac{1}{2}x_m \in M$ because M is convex but also we have that

$$\left\| \frac{x_n + x_m}{2} \right\| \geq d = \inf_{x\in M} \|x\|$$

Hence

$$\|x_n + x_m\| \geq 2d$$

On the other hand, by the parallelogram equality we get that

$$\begin{aligned} \|x_n - x_m\|^2 &= -\|x_n + x_m\|^2 + 2(\|x_n\|^2 + \|x_m\|^2) \\ &\leq -(2d)^2 + 2(\|x_n\|^2 + \|x_m\|^2) \end{aligned}$$

But since $\|x_n\| \to d$ as $n, m \to \infty$ then $\|x_n - x_m\| \to 0$. Therefore (x_n) is Cauchy, but this also implies that (x_n) converges because $\overline{M}$ is a subset of a Hilbert space. □

Proof. **5** Let $X = \mathbb{R}^2$ then

(a) Let $M = \{x\}$ where $x = (\xi_1, \xi_2) \neq 0$ then $M^\perp$ by definition is

$$M^\perp = \{x' \in X \mid x' \perp M\} = \{x' \in X \mid \langle x', x \rangle = 0\}$$

Let $x' = (-\xi_2, \xi_1) \in X$ then we see that

$$\langle x', x \rangle = -\xi_1\xi_2 + \xi_2\xi_1 = 0$$

So $(-\xi_2, \xi_1) \in M^\perp$ but $(-\alpha\xi_2, \alpha\xi_1)$ for $\alpha \in \mathbb{R}$ is also in $M^\perp$.
Therefore $M^\perp = \{(-\alpha\xi_2, \alpha\xi_1) \in X \mid \alpha \in \mathbb{R}\}$.

(b) Let $M = \{x_1, x_2\}$ where x_1 and x_2 are linearly independent then

$$\begin{aligned} M^\perp &= \{x' \in X \mid x' \perp M\} \\ &= \{x' \in X \mid \langle x', x_1 \rangle = 0 \text{ and } \langle x', x_2 \rangle = 0\} \end{aligned}$$

Suppose $x' \in X$ satisfy $\langle x', x_1 \rangle = 0$ and $\langle x', x_2 \rangle = 0$, then we have that

$$\langle x', x_1 \rangle = \eta_1\xi_{11} + \eta_2\xi_{12} = 0$$

and

$$\langle x', x_2 \rangle = \eta_1\xi_{21} + \eta_2\xi_{22} = 0$$

Where we assumed that $x_1 = (\xi_{11}, \xi_{12})$, $x_2 = (\xi_{21}, \xi_{22})$ and $x' = (\eta_1, \eta_2)$. But since x_1 and x_2 are linearly independent then the equations

$$\begin{aligned} c_1\xi_{11} + c_2\xi_{21} = 0 \\ c_1\xi_{12} + c_2\xi_{22} = 0 \end{aligned}$$

have only one solution, $c_1 = c_2 = 0$.
Therefore must be that $x' = 0$ and hence

$$M^\perp = \{0\}$$

$\square$

Proof. **6**
Let $Y = \{x \mid x = (\xi_j) \in l^2, \xi_{2n} = 0, n \in \mathbb{N}\}$ we want to show that Y is a closed subspace of l^2.
Let $(x_j) \subset Y$ and suppose it converges to some $x = (\eta_j) \notin Y$ we want to arrive at a contradiction, then given $\epsilon > 0$ there is some $N \in \mathbb{N}$ such that when $n \geq N$ we have that $\|x_n - x\| < \epsilon$, also, we see that

$$\|x_n - x\| = \sqrt{\sum_{j=1}^{\infty} |\xi_j - \eta_j|^2} = \sqrt{\sum_{j=1}^{\infty} |\xi_{2j+1} - \eta_{2j+1}|^2 + \sum_{j=1}^{\infty} |\eta_{2j}|^2}$$

Since the first term on the RHS is non-negative then

$$\|x_n - x\| \geq \sqrt{\sum_{j=1}^{\infty} |\eta_{2j}|^2}$$

So we have a lower bound on $\|x_n - x\|$ but $\|x_n - x\| \to 0$, a contradiction. Therefore, must be that $\eta_{2j} = 0$ so $x \in Y$ and hence Y is closed.

Given $x \in Y$, to find $Y^{\perp}$ we need to find all elements $z \in l^2$ such that $\langle x, z \rangle = 0$. We see that

$$\langle x, z \rangle = \sum_{j=1}^{\infty} \xi_j \bar{\eta}_j = \sum_{j=1}^{\infty} \xi_{2j+1} \bar{\eta}_{2j+1} + \sum_{j=1}^{\infty} \xi_{2j} \bar{\eta}_{2j} = \sum_{j=1}^{\infty} \xi_{2j+1} \bar{\eta}_{2j+1}$$

So for $\langle x, z \rangle = 0$ then must be that $\bar{\eta}_{2j+1} = 0$ because it must hold for every $x \in Y$. Therefore

$$Y^{\perp} = \{x \mid x = (\eta_j) \in l^2, \eta_{2n+1} = 0, n \in \mathbb{N}\}$$

Finally, let $Y = \text{span}\{e_1, ..., e_n\} \subset l^2$ where $e_j = (\delta_{jk})$.
We see that an element $x \in Y$ can be written as

$$x = \alpha_1 e_1 + ... + \alpha_n e_n = \Big(\alpha_1, \alpha_2, ..., \alpha_n, 0, 0, ...\Big)$$

Where $\alpha_1, ..., \alpha_n \in \mathbb{C}$.
Let then $z \in l^2$, we see that

$$\langle x, z \rangle = \alpha_1 \bar{\eta}_1 + \alpha_2 \bar{\eta}_2 + ... + \alpha_n \bar{\eta}_n + 0 \cdot \bar{\eta}_{n+1} + 0 \cdot \bar{\eta}_{n+2} + ...$$

So for $\langle x, z \rangle = 0$ must be that $\eta_j = 0$ for $1 \leq j \leq n$ and hence

$$Y^{\perp} = \{x \mid x = (\xi_j) \in l^2 \text{ such that } \xi_j = 0 \text{ for } 1 \leq j \leq n\}$$

□

Proof. **7**
Let A and $A \subset B$ be nonempty subsets of an inner product space X

(a) Let $a \in A$ and $a' \in A^\perp$ we see that $\langle a, a' \rangle = 0$ hence $a \perp A^\perp$, also by definition we know that

$$A^{\perp\perp} = \{x \in X \mid x \perp A^\perp\}$$

Then $a \in A^{\perp\perp}$ and since a was arbitrary we have that

$$A \subset A^{\perp\perp}$$

(b) Let $b \in B$, $a \in A$ and $b' \in B^\perp$, by definition $b' \perp B$ hence $\langle b, b' \rangle = 0$ but since a is also in B then must be that $\langle a, b' \rangle = 0$ and since a and b were arbitrary then this implies that $b' \perp A$ and therefore

$$B^\perp \subset A^\perp$$

(c) From part (a) we have that $A \subset A^{\perp\perp}$ and from part (b) taking $B = A^{\perp\perp}$ we have that

$$B^\perp = A^{\perp\perp\perp} \subset A^\perp$$

Also, from part (a) if we take A as $A^\perp$ following the same steps we get that

$$A^\perp \subset A^{\perp\perp\perp}$$

Therefore joining both results we get that $A^{\perp\perp\perp} = A^\perp$.

$\square$

Proof. **9**
Let Y be a subspace of a Hilbert space H

($\Rightarrow$) Suppose Y is closed in H then by Lemma 3.3-6 we get that $Y = Y^{\perp\perp}$.

($\Leftarrow$) Suppose $Y = Y^{\perp\perp}$. We prove first that $Y^{\perp}$ is closed in H.
Let $(x_n) \in Y^{\perp}$ be a convergent sequence, that converges to $x \in H$.
Let also, $y \in Y$ then we know that $\langle x_n, y \rangle = 0$, then by continuity of the inner product we have that

$$\langle x_n, y \rangle \to \langle x, y \rangle$$

but $\langle x_n, y \rangle = 0$ then $\langle x, y \rangle = 0$ and hence $x \in Y^{\perp}$. So $Y^{\perp}$ is closed.
We see that $Y^{\perp}$ is a subspace of H since if $y_1', y_2' \in Y^{\perp}$ and $x \in Y$ then

$$\left\langle \alpha y_1' + \beta y_2', x \right\rangle = \alpha \left\langle y_1', x \right\rangle + \beta \left\langle y_2', x \right\rangle = 0$$

So following the same logic we used to prove $Y^{\perp}$ is closed assuming Y is a subspace of H we get that $Y^{\perp\perp}$ is closed but $Y = Y^{\perp\perp}$.
Therefore Y is closed.

$\square$

Proof. **10**
Let $M \neq \emptyset$ be any subset of a Hilbert space H.
Let us notice that $M^{\perp\perp}$ is a closed subspace of H so $M^{\perp\perp}$ is a Hilbert space and hence without loss of generality we may assume that M is a subset of $H = M^{\perp\perp}$.
Let us take $x, y \in H = M^{\perp\perp}$ then $\langle x, y\rangle \neq 0$ because otherwise either x or y are not in $M^{\perp\perp}$ so the only way where $\langle x, y\rangle = 0$ is that $x = y = 0$ and hence $M^{\perp} = \{0\}$.
Then by theorem 3.3-7 we have that the span of M is dense in $M^{\perp\perp}$ but then

$$M \subset \overline{\text{span}(M)} = M^{\perp\perp}$$

Therefore $M^{\perp\perp}$ is the smallest closed subspace containing M. $\square$

3.4 - Orthonormal Sets and Sequences

Proof. **1.** Let X be an inner product space of finite dimension n, then, X contains a lineraly independent set of n vectors (a basis), let us call them $B' = \{b'_1, ..., b'_n\}$.
We want to define $B = \{b_1, ..., b_n\}$ such that the elements are orthonormal. Let us define $b_1 \in X$ such that

$$b_1 = \frac{1}{\|b'_1\|} \cdot b'_1$$

We see that we can write b'_2 as

$$b'_2 = \left\langle b'_2, b_1 \right\rangle b_1 + v_2$$

where v_2 is non-zero since the elements of B' are linearly independent, then

$$v_2 = b'_2 - \left\langle b'_2, b_1 \right\rangle b_1$$

We see that

$$\begin{aligned}
\langle b_1, v_2 \rangle &= \left\langle b_1, b'_2 - \left\langle b'_2, b_1 \right\rangle b_1 \right\rangle \\
&= \left\langle b_1, b'_2 \right\rangle - \left\langle b_1, \left\langle b'_2, b_1 \right\rangle b_1 \right\rangle \\
&= \left\langle b_1, b'_2 \right\rangle - \overline{\left\langle b'_2, b_1 \right\rangle} \left\langle b_1, b_1 \right\rangle \\
&= \left\langle b_1, b'_2 \right\rangle - \left\langle b_1, b'_2 \right\rangle \frac{1}{\|b'_1\|^2} \left\langle b'_1, b'_1 \right\rangle \\
&= \left\langle b_1, b'_2 \right\rangle - \left\langle b_1, b'_2 \right\rangle \frac{\|b'_1\|^2}{\|b'_1\|^2} \\
&= \left\langle b_1, b'_2 \right\rangle - \left\langle b_1, b'_2 \right\rangle \\
&= 0
\end{aligned}$$

Hence, we see that they are orthogonal. Then we define b_2 as

$$b_2 = \frac{1}{\|v_2\|} \cdot v_2$$

Now, to define b_3 we note that there is v_3 such that

$$v_3 = b'_3 - \left\langle b'_3, b_1 \right\rangle b_1 - \left\langle b'_3, b_2 \right\rangle b_2$$

We see that v_3 is non-zero since the elements of B' are linearly independent. Also, $v_3 \perp b_1$ and $v_3 \perp b_2$, then we define b_3 as

$$b_3 = \frac{1}{\|v_3\|} \cdot v_3$$

We can carry on this process n times until we find

$$v_n = b'_n - \sum_{i=1}^{n-1} \left\langle b'_n, b_i \right\rangle b_i$$

From which we define b_n as

$$b_n = \frac{1}{\|v_n\|} \cdot v_n$$

Therefore we have found a set B that is orthogonal pairwise, where each element has norm equal to 1, so B is an orthonormal set.
Finally, because of Lemma 3.4-2 we know that B is linearly independent so B is a basis for X as well. $\square$

Proof. **2.** Equation (12*) states that

$$\sum_{k=1}^{n} |\langle x, e_k \rangle|^2 \leq \|x\|^2$$

Where $x \in \mathbb{R}^r$ and $r \geq n$. We know from equation (9) that

$$\|y\|^2 = \sum_{k=1}^{n} |\langle x, e_k \rangle|^2$$

So y is the projection of x onto a set of orthonormal vectors in $\mathbb{R}^n$ hence (12*) is telling us geometrically that the length of a projection vector y is less than or equal to the length of x. □

Proof. **3.** Let $x, y \in X$ where X is an inner product space. Let us take an orthonormal set $\{e_1, ..., e_n\}$ such that

$$e_1 = \frac{y}{\|y\|}$$

Then we see that

$$\begin{aligned}
\sum_{k=1}^{n} |\langle x, e_k \rangle|^2 &= |\langle x, e_1 \rangle|^2 + \sum_{k=2}^{n} |\langle x, e_k \rangle|^2 \\
&= \left| \left\langle x, \frac{y}{\|y\|} \right\rangle \right|^2 + \sum_{k=2}^{n} |\langle x, e_k \rangle|^2 \\
&= \frac{|\langle x, y \rangle|^2}{\|y\|^2} + \sum_{k=2}^{n} |\langle x, e_k \rangle|^2
\end{aligned}$$

We note that

$$\frac{|\langle x, y \rangle|^2}{\|y\|^2} \leq \frac{|\langle x, y \rangle|^2}{\|y\|^2} + \sum_{k=2}^{n} |\langle x, e_k \rangle|^2$$

Now, applying (12*) to x we get that

$$\frac{|\langle x, y \rangle|^2}{\|y\|^2} \leq \frac{|\langle x, y \rangle|^2}{\|y\|^2} + \sum_{k=2}^{n} |\langle x, e_k \rangle|^2 = \sum_{k=1}^{n} |\langle x, e_k \rangle|^2 \leq \|x\|^2$$

Hence

$$|\langle x, y \rangle|^2 \leq \|x\|^2 \|y\|^2$$

And taking the square root we get Schwarz inequality

$$|\langle x, y \rangle| \leq \|x\| \|y\|$$

$\square$

Proof. **4.** Let us consider the orthonormal sequence

$$\begin{aligned} e_1 &= (0, 1, 0, 0, ...) \\ e_2 &= (0, 0, 1, 0, ...) \\ e_3 &= (0, 0, 0, 1, ...) \\ &\vdots \end{aligned}$$

And let us consider the sequence

$$x = (1, 0, 0, 0, ...) \in l^2$$

Then we see that

$$\sum_{k=1}^{\infty} |\langle x, e_k \rangle|^2 = 0$$

But

$$\|x\|^2 = 1^2 + 0 + 0 + ... = 1$$

Therefore the strict inequality holds in this case

$$\sum_{k=1}^{\infty} |\langle x, e_k \rangle|^2 = 0 < \|x\|^2 = 1$$

□

Proof. **5.** Let (e_k) be an orthonormal sequence in an inner product space X, let also $x \in X$ and

$$y = \sum_{k=1}^{n} \alpha_k e_k \qquad \alpha_k = \langle x, e_k \rangle$$

We want to show that $x-y$ is orthogonal to the subspace $Y_n = \text{span}\{e_1, ..., e_n\}$. Let us consider some $y' \in Y_n$ defined as

$$y' = \sum_{k=1}^{n} \left\langle y', e_k \right\rangle e_k$$

Then

$$\begin{aligned}
\left\langle x - y, y' \right\rangle &= \left\langle x, y' \right\rangle - \left\langle y, y' \right\rangle \\
&= \left\langle x, \sum_{k=1}^{n} \left\langle y', e_k \right\rangle e_k \right\rangle - \left\langle \sum_{k=1}^{n} \langle x, e_k \rangle e_k, \sum_{j=1}^{n} \left\langle y', e_j \right\rangle e_j \right\rangle \\
&= \sum_{k=1}^{n} \overline{\langle y', e_k \rangle} \langle x, e_k \rangle - \sum_{k=1}^{n} \langle x, e_k \rangle \left\langle e_k, \sum_{j=1}^{n} \left\langle y', e_j \right\rangle e_j \right\rangle \\
&= \sum_{k=1}^{n} \overline{\langle y', e_k \rangle} \langle x, e_k \rangle - \sum_{k=1}^{n} \langle x, e_k \rangle \sum_{j=1}^{n} \overline{\langle y', e_j \rangle} \langle e_k, e_j \rangle
\end{aligned}$$

Since $\langle e_k, e_j \rangle = 1$ only if $k = j$ otherwise it's 0 then we can write that

$$\left\langle x - y, y' \right\rangle = \sum_{k=1}^{n} \overline{\langle y', e_k \rangle} \langle x, e_k \rangle - \sum_{k=1}^{n} \langle x, e_k \rangle \overline{\langle y', e_k \rangle} = 0$$

Therefore since y' was arbitrary we see that $x - y$ is orthogonal to Y_n. $\square$

Proof. **6.** Let $\{e_1, ..., e_n\}$ be an orthonormal set in an inner product space X, where n is fixed. Let $x \in X$ and let $y = \beta_1 e_1 + ... + \beta_n e_n$. We want to show that $\|x - y\|$ is minimum if and only if $\beta_j = \langle x, e_j \rangle$ where $j = 1, ..., n$.

($\Rightarrow$) Suppose $\|x - y\|$ is minimum then we note that

$$x - y = x - \sum_{j=1}^{n} \beta_j e_j = \left(x - \sum_{i=1}^{n} \langle x, e_i \rangle e_i\right) + \left(\sum_{k=1}^{n} \langle x, e_k \rangle e_k - \sum_{j=1}^{n} \beta_j e_j\right)$$

Then we see that

$$\begin{aligned}
&\left\langle x - \sum_{i=1}^{n} \langle x, e_i \rangle e_i, \sum_{k=1}^{n} \langle x, e_k \rangle e_k - \sum_{j=1}^{n} \beta_j e_j \right\rangle \\
&= \left\langle x, \sum_{k=1}^{n} \langle x, e_k \rangle e_k \right\rangle - \left\langle x, \sum_{j=1}^{n} \beta_j e_j \right\rangle - \left\langle \sum_{i=1}^{n} \langle x, e_i \rangle e_i, \sum_{k=1}^{n} \langle x, e_k \rangle e_k \right\rangle \\
&\quad + \left\langle \sum_{i=1}^{n} \langle x, e_i \rangle e_i, \sum_{j=1}^{n} \beta_j e_j \right\rangle \\
&= \sum_{k=1}^{n} \overline{\langle x, e_k \rangle} \langle x, e_k \rangle - \sum_{j=1}^{n} \overline{\beta_j} \langle x, e_j \rangle - \sum_{i=1}^{n} \langle x, e_i \rangle \sum_{k=1}^{n} \overline{\langle x, e_k \rangle} \langle e_i, e_k \rangle \\
&\quad + \sum_{i=1}^{n} \langle x, e_i \rangle \sum_{j=1}^{n} \overline{\beta_j} \langle e_i, e_j \rangle \\
&= \sum_{k=1}^{n} \overline{\langle x, e_k \rangle} \langle x, e_k \rangle - \sum_{j=1}^{n} \overline{\beta_j} \langle x, e_j \rangle - \sum_{i=1}^{n} \langle x, e_i \rangle \overline{\langle x, e_i \rangle} + \sum_{i=1}^{n} \langle x, e_i \rangle \overline{\beta_i} \\
&= 0
\end{aligned}$$

So the vectors are orthogonal, then by the Pythagorean realation we have that

$$\|x - \sum_{j=1}^{n} \beta_j e_j\|^2 = \|x - \sum_{i=1}^{n} \langle x, e_i \rangle e_i\|^2 + \|\sum_{k=1}^{n} \langle x, e_k \rangle e_k - \sum_{j=1}^{n} \beta_j e_j\|^2$$

We see the last term is

$$\begin{aligned}
\|\sum_{k=1}^{n} (\langle x, e_k \rangle - \beta_k) e_k\|^2 &= \left\langle \sum_{k=1}^{n} (\langle x, e_k \rangle - \beta_k) e_k, \sum_{j=1}^{n} (\langle x, e_j \rangle - \beta_j) e_j \right\rangle \\
&= \sum_{k=1}^{n} (\langle x, e_k \rangle - \beta_k) \sum_{j=1}^{n} (\langle x, e_j \rangle - \beta_j) \langle e_k, e_j \rangle \\
&= \sum_{k=1}^{n} |\langle x, e_k \rangle - \beta_k|^2
\end{aligned}$$

Therefore since the first term, $\|x - \sum_{i=1}^{n} \langle x, e_i \rangle e_i\|^2$, is constant, and we said that $\|x - y\|^2$ is mininum then must be that

$$\sum_{k=1}^{n} | \langle x, e_k \rangle - \beta_k|^2 = 0$$

And that can only happen if $\beta_k = \langle x, e_k \rangle$ for all $k \in \{1, ..., n\}$.

($\Leftarrow$) Suppose now that $\beta_j = \langle x, e_j \rangle$ for all $j \in \{1, ..., n\}$ then

$$\|x - \sum_{j=1}^{n} \beta_j e_j\|^2 = \|x - \sum_{i=1}^{n} \langle x, e_i \rangle e_i\|^2 + \sum_{j=1}^{n} | \langle x, e_j \rangle - \beta_j|^2 = \|x - \sum_{i=1}^{n} \langle x, e_i \rangle e_i\|^2$$

Since $\|x - \sum_{i=1}^{n} \langle x, e_i \rangle e_i\|^2$ is a constant then $\|x - y\|$ cannot take a smaller value, i.e. it's the minimum when $\beta_j = \langle x, e_j \rangle$.

$\square$

Proof. **7.** Let (e_j) be any orthonormal sequence in an inner product space X and let $x, y \in X$.
Let us consider the Cauchy-Schwartz inequality over the sequences $(\langle x, e_k \rangle)$ and $(\langle y, e_k \rangle)$, then

$$\sum_{k=1}^{\infty} | \langle x, e_k \rangle \langle y, e_k \rangle | \leq \sqrt{\sum_{j=1}^{\infty} | \langle x, e_j \rangle |^2} \sqrt{\sum_{m=1}^{\infty} | \langle y, e_m \rangle |^2}$$

We know that $\sum_{j=1}^{\infty} | \langle x, e_j \rangle |^2 \leq \|x\|^2$ because of (12) hence

$$\sum_{k=1}^{\infty} | \langle x, e_k \rangle \langle y, e_k \rangle | \leq \sqrt{\|x\|^2} \sqrt{\|y\|^2} = \|x\| \|y\|$$

$\square$

Proof. **8.** Let $x \in X$ where X is an inner product space and let n_m be the number of $\langle x, e_k \rangle$ such that $| \langle x, e_k \rangle | > 1/m$.
From equation (12) we know that

$$\sum_{k=1}^{\infty} | \langle x, e_k \rangle |^2 \leq \|x\|^2$$

So we see that

$$\frac{n_m}{m^2} < \sum_{k=1}^{\infty} | \langle x, e_k \rangle |^2 \leq \|x\|^2$$

Therefore, must be that

$$n_m < m^2 \|x\|^2$$

$\square$

Proof. **9.** Let the sequence $(x_0, x_1, x_2, ...)$ where $x_j(t) = t^j$ on the interval $[-1, 1]$ where

$$\langle x, y\rangle = \int_{-1}^{1} x(t)y(t)\ dt$$

To orthonormalize the first three terms of the sequence we follow the Gram-Schmidt process. Let us compute first $\|x_0\|$ as follows

$$\|x_0\| = \sqrt{\langle x_0, x_0\rangle} = \sqrt{\int_{-1}^{1} x_0(t)x_0(t)\ dt} = \sqrt{\int_{-1}^{1} t^0 t^0\ dt} = \sqrt{\int_{-1}^{1} dt} = \sqrt{2}$$

So the first element of our orthonormalized sequence is

$$e_0 = \frac{1}{\|x_0\|}x_0 = \frac{1}{\sqrt{2}} \cdot t^0 = \frac{1}{\sqrt{2}}$$

Now we compute v_1 as

$$\begin{aligned} v_1 &= x_1 - \langle x_1, e_0\rangle e_0 \\ &= t - \frac{1}{\sqrt{2}} \cdot \int_{-1}^{1} t \cdot \frac{1}{\sqrt{2}}\ dt \\ &= t - \frac{1}{2}\int_{-1}^{1} t\ dt \\ &= t - \frac{1}{2}\left[\frac{t^2}{2}\right]_{-1}^{1} \\ &= t \end{aligned}$$

So the norm of v_1 is

$$\|v_1\| = \sqrt{\int_{-1}^{1} t^2\ dt} = \sqrt{\frac{2}{3}}$$

Then, the second element of the sequence is given by

$$e_1 = \frac{1}{\|v_1\|}v_1 = \sqrt{\frac{3}{2}}t$$

Finally, we compute v_2 as

$$\begin{aligned} v_2 &= x_2 - \langle x_2, e_1\rangle e_1 - \langle x_2, e_0\rangle e_0 \\ &= t^2 - \sqrt{\frac{3}{2}}t \cdot \int_{-1}^{1} t^2 \cdot \sqrt{\frac{3}{2}}t\ dt - \frac{1}{\sqrt{2}} \cdot \int_{-1}^{1} t^2 \cdot \frac{1}{\sqrt{2}}\ dt \\ &= t^2 - \frac{3}{2}t \cdot \int_{-1}^{1} t^3\ dt - \frac{1}{2}\int_{-1}^{1} t^2\ dt \\ &= t^2 - \frac{1}{3} \end{aligned}$$

Hence

$$\begin{aligned}
\|v_2\| &= \sqrt{\int_{-1}^{1}\left(t^2-\frac{1}{3}\right)^2 dt} \\
&= \sqrt{\int_{-1}^{1}\left(t^4-\frac{2t^2}{3}+\frac{1}{9}\right) dt} \\
&= \sqrt{\frac{2}{5}-\frac{4}{9}+\frac{2}{9}} \\
&= \sqrt{\frac{8}{45}}
\end{aligned}$$

so the third element of the sequence is

$$e_2 = \frac{1}{\|v_2\|}v_2 = \sqrt{\frac{45}{8}}\left(t^2-\frac{1}{3}\right)$$

Just as a small sanity check we compute $\|e_2\|$ as follows

$$\begin{aligned}
\|e_2\| &= \sqrt{\int_{-1}^{1}\frac{45}{8}\left(t^2-\frac{1}{3}\right)^2 dt} \\
&= \sqrt{\frac{45}{8}}\sqrt{\int_{-1}^{1}\left(t^2-\frac{1}{3}\right)^2 dt} \\
&= \sqrt{\frac{45}{8}}\|v_2\| \\
&= \sqrt{\frac{45}{8}}\sqrt{\frac{8}{45}} \\
&= 1
\end{aligned}$$

$\square$

Proof. **1.** Suppose (6) converges to x i.e.

$$\sum_{k=1}^{\infty} \alpha_k e_k = x$$

Then by Theorem 3.5-2 we know that (7)

$$\sum_{k=1}^{\infty} |\alpha_k|^2$$

converges.
Let us take the squared norm of x then we have that

$$\|x\|^2 = \left\| \sum_{k=1}^{\infty} \alpha_k e_k \right\|^2$$

But since every $\alpha_k e_k$ is orthonormal then by the Pythagorean relation we have that

$$\left\| \sum_{k=1}^{\infty} \alpha_k e_k \right\|^2 = \sum_{k=1}^{\infty} \|\alpha_k e_k\|^2$$

Therefore

$$\|x\|^2 = \left\| \sum_{k=1}^{\infty} \alpha_k e_k \right\|^2 = \sum_{k=1}^{\infty} \|\alpha_k e_k\|^2 = \sum_{k=1}^{\infty} |\alpha_k|^2 \|e_k\|^2 = \sum_{k=1}^{\infty} |\alpha_k|^2$$

$\square$

Proof. **3.** Let us consider the orthonormal sequence

$$\begin{aligned} e_1 &= (0,1,0,0,...) \\ e_2 &= (0,0,1,0,...) \\ e_3 &= (0,0,0,1,...) \\ &\vdots \end{aligned}$$

And let us consider the sequence

$$x = (1,0,0,0,...) \in l^2$$

Then we see that

$$\sum_{k=1}^{\infty} \langle x, e_k \rangle \, e_k = (0,0,0,0,...) \neq (1,0,0,0,...) = x$$

□

Proof. **4.**
Let $(x_j) \subset X$ such that $\|x_1\| + \|x_2\| + ...$ converges.
We want to prove that (s_n) is Cauchy where $s_n = \sum_{i=1}^{n} x_i$. Taking $n', m' \in \mathbb{N}$ such that $n' < m'$, we see that

$$\left\| \sum_{i=1}^{n'} x_i - \sum_{j=1}^{m'} x_j \right\| = \left\| \sum_{i=n'+1}^{m'} x_i \right\| \leq \sum_{i=n'+1}^{m'} \|x_i\|$$

On the other hand, since $\sum_i \|x_i\|$ converges then, it's Cauchy, so, given $\epsilon > 0$ there is $N \in \mathbb{N}$ such that when $n, m \geq N$ we have that

$$\left| \sum_{i=1}^{n} \|x_i\| - \sum_{j=1}^{m} \|x_j\| \right| < \epsilon$$

We may assume that $n < m$, then

$$\left| \sum_{i=n+1}^{m} \|x_i\| \right| = \sum_{i=n+1}^{m} \|x_i\| < \epsilon$$

Hence, taking $n' = n$ and $m' = m$ we see that

$$\left\| \sum_{i=1}^{n} x_i - \sum_{j=1}^{m} x_j \right\| = \left\| \sum_{i=n+1}^{m} x_i \right\| \leq \sum_{i=n+1}^{m} \|x_i\| < \epsilon$$

Therefore (s_n) is Cauchy. $\square$

Proof. **5.**
From problem **4.** we know that if X is an inner product space and $\sum_i \|x_i\|$ converges then the sequence (s_n) with $s_n = \sum_{i=1}^n x_i$ is Cauchy.
If instead we take (s_n) in a Hilbert space H, the space is complete so all Cauchy sequences converge, hence the sequence of partial sums (s_n) converge to some limit s, so we have that

$$\sum_{i=1}^{\infty} x_i = \lim_{n\to\infty} \sum_{i=1}^{n} x_i = \lim_{n\to\infty} s_n = s$$

Which implies that $\sum_{i=1}^{\infty} x_i$ converges. $\square$

Proof. **6.**
Let (e_j) be an orthonormal sequence in Hilbert space H and let

$$x = \sum_{j=1}^{\infty} \alpha_j e_j \qquad y = \sum_{j=1}^{\infty} \beta_j e_j$$

Let us define the partial sums sequences

$$x_n = \sum_{j=1}^{n} \alpha_j e_j \qquad y_n = \sum_{j=1}^{n} \beta_j e_j$$

we see that $x_n \to x$ and $y_n \to y$ as $n \to \infty$.
Then

$$\langle x_n, y_n \rangle = \left\langle \sum_{j=1}^{n} \alpha_j e_j, \sum_{i=1}^{n} \beta_i e_i \right\rangle = \sum_{j=1}^{n} \alpha_j \sum_{i=1}^{n} \overline{\beta_i} \langle e_j, e_i \rangle = \sum_{j=1}^{n} \alpha_j \overline{\beta_j}$$

Because $x_n \to x$, $y_n \to y$ and the continuity of the inner product we have that $\langle x_n, y_n \rangle \to \langle x, y \rangle$. Therefore

$$\langle x, y \rangle = \lim_{n\to\infty} \langle x_n, y_n \rangle = \lim_{n\to\infty} \sum_{j=1}^{n} \alpha_j \overline{\beta_j} = \sum_{j=1}^{\infty} \alpha_j \overline{\beta_j}$$

$\square$

Proof. **7.**
Let (e_k) be an orthonormal sequence in a Hilbert space H and let $x \in H$. We want to show that

$$y = \sum_{k=1}^{\infty} \langle x, e_k \rangle e_k$$

exists in H and that $x - y$ is orthogonal to every e_k.
Let us consider the partial sum sequence

$$y_n = \sum_{k=1}^{n} \langle x, e_k \rangle e_k$$

If we let $m < n$ then we see that

$$\begin{aligned}
\|y_n - y_m\|^2 &= \left\| \sum_{k=1}^{n} \langle x, e_k \rangle e_k - \sum_{j=1}^{m} \langle x, e_j \rangle e_j \right\|^2 \\
&= \left\| \sum_{k=m+1}^{n} \langle x, e_k \rangle e_k \right\|^2 \\
&= \sum_{k=m+1}^{n} \| \langle x, e_k \rangle e_k \|^2 \\
&= \sum_{k=m+1}^{n} | \langle x, e_k \rangle |^2
\end{aligned}$$

On the other hand, by Bessel inequality we know that $\sum_{k=1}^{\infty} | \langle x, e_k \rangle |^2$ converges so the tail of the series $\sum_{k=m+1}^{n} | \langle x, e_k \rangle |^2$ must tend to 0 as $n, m \to \infty$.
Therefore $\|y_n - y_m\|^2 \to 0$ as $n, m \to \infty$ and hence (y_n) is Cauchy and since $(y_n) \in H$ then (y_n) converges. So

$$y = \lim_{n \to \infty} \sum_{k=1}^{n} \langle x, e_k \rangle e_k = \sum_{k=1}^{\infty} \langle x, e_k \rangle e_k$$

exists in H.
Let us consider now the following

$$\begin{aligned}
\langle x - y_n, e_k \rangle &= \left\langle x - \sum_{j=1}^{n} \langle x, e_j \rangle e_j, e_k \right\rangle \\
&= \langle x, e_k \rangle - \left\langle \sum_{j=1}^{n} \langle x, e_j \rangle e_j, e_k \right\rangle \\
&= \langle x, e_k \rangle - \sum_{j=1}^{n} \langle x, e_j \rangle \langle e_j, e_k \rangle
\end{aligned}$$

Now taking the limit as $n \to \infty$ we get that

$$\begin{aligned}
\lim_{n\to\infty} \langle x - y_n, e_k \rangle &= \langle x, e_k \rangle - \lim_{n\to\infty} \sum_{j=1}^{n} \langle x, e_j \rangle \langle e_j, e_k \rangle \\
\langle x - y, e_k \rangle &= \langle x, e_k \rangle - \sum_{j=1}^{\infty} \langle x, e_j \rangle \langle e_j, e_k \rangle \\
\langle x - y, e_k \rangle &= \langle x, e_k \rangle - \langle x, e_k \rangle \\
\langle x - y, e_k \rangle &= 0
\end{aligned}$$

Where we used the continuity of the inner product, to show that as $n \to \infty$ we have that $\langle x - y_n, e_k \rangle \to \langle x - y, e_k \rangle$.
Therefore $x - y$ is orthogonal to every e_k.

$$\left\langle \sum_{j=1}^{\infty} \langle x, e_j \rangle e_j, e_k \right\rangle = \sum_{j=1}^{\infty} \langle x, e_j \rangle \langle e_j, e_k \rangle$$

□

Proof. **8.**
Let (e_k) be an orthonormal sequence in a Hilbert space H and let M be $M = \text{span}(e_k)$.

($\Rightarrow$) Let $x \in \bar{M}$ then x can be written as

$$x = \sum_{k=1}^{\infty} \alpha_k e_k$$

Where α_k are scalars. Since the summation converges to x then by Theorem 3.5-2 (b) we know the coefficients α_k are the Fourier coefficients $\langle x, e_k \rangle$ so x can be written as

$$x = \sum_{k=1}^{\infty} \langle x, e_k \rangle e_k$$

($\Leftarrow$) Let $x \in H$ such that it can be written as

$$x = \sum_{k=1}^{\infty} \langle x, e_k \rangle e_k = \sum_{k=1}^{\infty} \alpha_k e_k$$

Then x is an infinite linear combination of the elements of the orthonormal sequence (e_k) so must be that $x \in \bar{M}$.

$\square$

Proof. **9.**
Let (e_n) and $(\tilde{e}_n)$ be orthonormal sequences in a Hilbert space H and let $M_1 = \text{span}(e_n)$ and $M_2 = \text{span}(\tilde{e}_n)$.

($\Rightarrow$) Let $\bar{M}_1 = \bar{M}_2$. By problem 8 we know that if $x \in \bar{M}_1$ and $x \in H$ then it can be written as

$$x = \sum_{m=1}^{\infty} \alpha_m e_m = \sum_{m=1}^{\infty} \langle x, e_m \rangle e_m$$

We note that $e_n \in \bar{M}_2$ then we can write it as

$$e_n = \sum_{m=1}^{\infty} \langle e_n, \tilde{e}_m \rangle \tilde{e}_m = \sum_{m=1}^{\infty} \alpha_{nm} \tilde{e}_m$$

Where we defined $\alpha_{nm} = \langle e_n, \tilde{e}_m \rangle$.
In the same way, since $\tilde{e}_n \in M_1$ then we can write it as

$$\tilde{e}_n = \sum_{m=1}^{\infty} \langle \tilde{e}_n, e_m \rangle e_m = \sum_{m=1}^{\infty} \overline{\langle e_m, \tilde{e}_n \rangle} e_m = \sum_{m=1}^{\infty} \bar{\alpha}_{mn} e_m$$

($\Leftarrow$) Let us suppose that

$$e_n = \sum_{m=1}^{\infty} \alpha_{nm} \tilde{e}_m \quad \text{and} \quad \tilde{e}_n = \sum_{m=1}^{\infty} \bar{\alpha}_{mn} e_m$$

Where $\alpha_{nm} = \langle e_n, \tilde{e}_m \rangle$.
We see that e_n and $\tilde{e}_n$ can be written in the form of equation (6) then the orhtonormal sequence (e_n) is both in $\bar{M}_1$ and $\bar{M}_2$ and the orhtonormal sequence $(\tilde{e}_n)$ is also in $\bar{M}_1$ and $\bar{M}_2$.
Let us take $x \in \bar{M}_1$ then by problem 8 we know that x can be written as

$$x = \sum_{m=1}^{\infty} \langle x, e_m \rangle e_m$$

But since the orthonormal sequence (e_n) is also in $\bar{M}_2$ then this implies that $x \in \bar{M}_2$ and since x is arbitrary then

$$\bar{M}_1 \subset \bar{M}_2$$

The opposite can be proven if $x \in \bar{M}_2$ so we get that

$$\bar{M}_1 = \bar{M}_2$$

$\square$

3.6 - Total Orthonormal Sets and Sequences

Proof. **1.**
Let F be an orthonormal basis in an inner product space X. We know that

$$\overline{\operatorname{span}(F)} = X$$

so if $x \in X$ can be represented as a linear combination of elements of F then $x \in \operatorname{span}(F)$ but it could happen that $x \in X$ but it cannot be represented as a finite linear combination of elements in F.
For example, let us consider $X = l^2$ then if we take F as

$$F = \{(1,0,0,...),(0,1,0,...),(0,0,1,...),...\}$$

We see that $\overline{\operatorname{span}(F)} = l^2$, also, the sequence $(1, 1/2, 1/3, ...) \in l^2$ but is not in $\operatorname{span}(F)$ since it's infinite. □

Proof. **2.**
Suppose the orthogonal dimension of a hilbert space H is finite, then there is a total orthonormal set T with finite number of elements in it, let us suppose this number is n, but by Lemma 3.4-2 we know that orthonormal sets are linealy independent so inside H we have a set of n linearly independent elements.
Let us suppose that there is, for example, one more element that is linearly independent with respect to the n previous elements, we want to arrive at a contradiction. So, in total we have $n+1$ linearly independent elements, then, we can build an orthonormal set F of $n+1$ elements using the Gram-Schmidt process, since this new orthonormal set was built from the original total orthonormal set T then also, must be that $\operatorname{span}(F)$ is dense in H, but for this to happen by Lemma 3.3-7 must be that $\operatorname{span}(F)^{\perp} = \{0\}$ which implies that $\operatorname{span}(T)^{\perp} \neq \{0\}$ in the first place.
This is a contradiction and must be that there cannot be more than n linearly independent sets if the orthogonal dimension of H is n.
Therefore, the vector space H is of dimension n.
Now, suppose H is considered as a vector space is of dimension n (finite), then there is a linearly independent set with n elements in it. We can obtain an orthonormal set of n elements by applying the Gram-Schmidt process and we observe that all elements of H are in the span of the orthonormal set, so the orthonormal set is total, hence, the orthogonal dimension of H is n. □

Proof. **4.**
Let us consider first the case of a complex inner product space. From Parseval relation we note that

$$\|x-y\|^2 = \sum_k |\langle x-y, e_k\rangle|^2$$

Hence, from the LHS we see that

$$\|x-y\|^2 = \langle x-y, x-y\rangle = \langle x,x\rangle - \langle x,y\rangle - \langle y,x\rangle + \langle y,y\rangle$$

And, from the RHS we have that

$$\begin{aligned}
\sum_k |\langle x-y, e_k\rangle|^2 &= \sum_k \langle x-y, e_k\rangle \overline{\langle x-y, e_k\rangle} \\
&= \sum_k (\langle x, e_k\rangle - \langle y, e_k\rangle)(\overline{\langle x, e_k\rangle} - \overline{\langle y, e_k\rangle}) \\
&= \sum_k \langle x, e_k\rangle \overline{\langle x, e_k\rangle} - \langle y, e_k\rangle \overline{\langle x, e_k\rangle} - \langle x, e_k\rangle \overline{\langle y, e_k\rangle} + \langle y, e_k\rangle \overline{\langle y, e_k\rangle} \\
&= \sum_k |\langle x, e_k\rangle|^2 - \langle y, e_k\rangle \overline{\langle x, e_k\rangle} - \langle x, e_k\rangle \overline{\langle y, e_k\rangle} + |\langle y, e_k\rangle|^2 \\
&= \langle x,x\rangle + \langle y,y\rangle - \sum_k \langle y, e_k\rangle \overline{\langle x, e_k\rangle} + \langle x, e_k\rangle \overline{\langle y, e_k\rangle}
\end{aligned}$$

Joining both results we see that

$$\langle x,y\rangle + \langle y,x\rangle = \sum_k \langle y, e_k\rangle \overline{\langle x, e_k\rangle} + \langle x, e_k\rangle \overline{\langle y, e_k\rangle}$$

Now, let us consider

$$\|x-iy\|^2 = \sum_k |\langle x-iy, e_k\rangle|^2$$

From the LHS we see that

$$\begin{aligned}
\|x-iy\|^2 &= \langle x-iy, x-iy\rangle \\
&= \langle x,x\rangle - \langle x,iy\rangle - \langle iy,x\rangle + \langle iy,iy\rangle \\
&= \langle x,x\rangle + i\langle x,y\rangle - i\langle y,x\rangle + \langle y,y\rangle
\end{aligned}$$

And, from the RHS we have that

$$\begin{aligned}
\sum_k |\langle x-iy, e_k\rangle|^2 &= \sum_k \langle x-iy, e_k\rangle \overline{\langle x-iy, e_k\rangle} \\
&= \sum_k (\langle x, e_k\rangle - i\langle y, e_k\rangle)(\overline{\langle x, e_k\rangle} + i\overline{\langle y, e_k\rangle}) \\
&= \sum_k \langle x, e_k\rangle \overline{\langle x, e_k\rangle} - i\langle y, e_k\rangle \overline{\langle x, e_k\rangle} + i\langle x, e_k\rangle \overline{\langle y, e_k\rangle} + \langle y, e_k\rangle \overline{\langle y, e_k\rangle} \\
&= \sum_k |\langle x, e_k\rangle|^2 - i\langle y, e_k\rangle \overline{\langle x, e_k\rangle} + i\langle x, e_k\rangle \overline{\langle y, e_k\rangle} + |\langle y, e_k\rangle|^2 \\
&= \langle x,x\rangle + \langle y,y\rangle + i\sum_k \langle x, e_k\rangle \overline{\langle y, e_k\rangle} - \langle y, e_k\rangle \overline{\langle x, e_k\rangle}
\end{aligned}$$

Again, joining both results we see that

$$\langle x, y\rangle - \langle y, x\rangle = \sum_k \langle x, e_k\rangle \overline{\langle y, e_k\rangle} - \langle y, e_k\rangle \overline{\langle x, e_k\rangle}$$

Adding both result equations we get that

$$\begin{aligned}&\langle x, y\rangle + \langle y, x\rangle + \langle x, y\rangle - \langle y, x\rangle \\ &\quad = \sum_k \langle y, e_k\rangle \overline{\langle x, e_k\rangle} + \langle x, e_k\rangle \overline{\langle y, e_k\rangle} + \sum_k \langle x, e_k\rangle \overline{\langle y, e_k\rangle} - \langle y, e_k\rangle \overline{\langle x, e_k\rangle}\end{aligned}$$

Hence

$$\langle x, y\rangle = \sum_k \langle x, e_k\rangle \overline{\langle y, e_k\rangle}$$

Let us consider now the case of a real inner product space. From the polarization identity we know that

$$\langle x, y\rangle = \frac{1}{2}(\|x\|^2 + \|y\|^2 - \|x - y\|^2)$$

Hence

$$\begin{aligned}\langle x, y\rangle &= \frac{1}{2}(\langle x, x\rangle + \langle y, y\rangle - \langle x, x\rangle - \langle y, y\rangle + \sum_k \langle y, e_k\rangle \langle x, e_k\rangle + \langle x, e_k\rangle \langle y, e_k\rangle) \\ &= \frac{1}{2}(2\sum_k \langle x, e_k\rangle \langle y, e_k\rangle) \\ &= \sum_k \langle x, e_k\rangle \langle y, e_k\rangle\end{aligned}$$

Where we used the previous result we got for $\|x - y\|^2$ and that $\langle x, e_k\rangle = \langle e_k, x\rangle$ in a real inner product space. $\square$

Proof. **5.**
Let (e_κ), with $\kappa \in I$ be an orthonormal family in a Hilbert space H.

(⇒) Let us suppose (e_κ) is total. Since Parseval relation

$$\sum_k |\langle x, e_k\rangle|^2 = \|x\|^2$$

Holds for any total orthonormal set M in a Hilbert space H and we derived the equation

$$\langle x, y\rangle = \sum_k \langle x, e_k\rangle \overline{\langle y, e_k\rangle}$$

Using the Parseval relation then it must also hold for the orthonormal family (e_κ).

(⇐) Let us suppose that for every $x, y \in H$ the following equation holds

$$\langle x, y\rangle = \sum_\kappa \langle x, e_\kappa\rangle \overline{\langle y, e_\kappa\rangle}$$

For the orthonormal family (e_κ). Taking $y = x$ we get that

$$\|x\|^2 = \langle x, x\rangle = \sum_\kappa \langle x, e_\kappa\rangle \overline{\langle x, e_\kappa\rangle} = \sum_\kappa |\langle x, e_\kappa\rangle|^2$$

Then Parseval relation also holds and by Theorem 3.6-3 then must be that the orthonormal family (e_κ) is total.

□

Proof. **6.**
Let H be a separable Hilbert space and $M = \{x_1, x_2, x_3, ...\}$ a countable dense subset of H. Then, let us select

$$e_1 = \frac{x_1}{\|x_1\|}$$

To select e_2 we check if x_2 is in $\text{span}(e_1)$, if it is there then we drop it, if it's not then we define $v_2 = x_2 - \langle x_2, e_1 \rangle e_1$ and we select

$$e_2 = \frac{v_2}{\|v_2\|}$$

To select e_3 we check if x_3 is in $\text{span}(\{e_1, e_2\})$, if it is there then we drop it, if it's not then we define $v_3 = x_3 - \langle x_3, e_1 \rangle e_1 - \langle x_3, e_2 \rangle e_2$ and we select

$$e_3 = \frac{v_3}{\|v_3\|}$$

i.e. we follow the Gram-Schmidt process dropping some elements if necessary, to get an orthonormal sequence $(e_j) \subseteq M$.
Since the elements we dropped are in some span then they can be written as a linear combination of elements of (e_j), so $M \subseteq \text{span}(e_j)$ but also we know that M is a dense subset of H so must be that $\text{span}(e_j)$ is also dense in H. Therefore (e_j) is a total orthonormal sequence obtained from M. $\square$

Proof. **7.**
Let H be a separable Hilbert space, then there is M a countable dense subset in H. As we showed before, using the Gram-Schmidt process we can find a total orthonormal sequence in H. □

Proof. **8.**
Let $F = (e_j)$ be an orthonormal sequence in a separable Hilbert space H.
We know there exists $D = \{d_1, d_2, d_3, ...\}$, a countable dense subset of H, because H is separable.
Let us take d_1, if d_1 is linearly independent from the rest of the elements in D we keep it.
Next, we compute

$$v_1 = d_1 - \sum_{k=1}^{\infty} \langle d_1, e_k \rangle e_k$$

and

$$x_1 = \frac{v_1}{\|v_1\|}$$

i.e. we follow the Gram-Schmidt process to orthonormalize d_1.
Finally, we add x_1 to a new orthogonal sequence $\tilde{F}$ defined as

$$\tilde{F} = \{x_1, x_2, ..., e_1, e_2, ...\}$$

To get each x_i we follow the same process for each d_i. If d_i is not linearly independent we drop it and continue.
Following this process we see that $\tilde{F}$ is an orthonormal sequence, and since we started from a countable dense subset D, then $\text{span}(\tilde{F})$ must also be dense in H, since we only dropped the elements that were linearly dependent to the other elements.
Therefore $\tilde{F}$ is a total orthonormal sequence that contains F. □

Proof. **9.**
Let M be a total set in an inner product space X and let $\langle v, x\rangle = \langle w, x\rangle$ for all $x \in M$. We note that

$$\begin{aligned} \langle v, x\rangle &= \langle w, x\rangle \\ \langle v, x\rangle - \langle w, x\rangle &= 0 \\ \langle v - w, x\rangle &= 0 \end{aligned}$$

This implies that $v - w \perp x$ but because Theorem 3.6-2, there is no element in X which is orthogonal to every element in M so must be that $v - w = 0$ and therefore $v = w$. □

Proof. **10.**
Let M be a subset of a Hilbert space H and let $v, w \in H$. Suppose that $\langle v, x \rangle = \langle w, x \rangle$ for all $x \in M$ implies that $v = w$.
If M is not total then there is some $y \in H$, $y \neq 0$ such that $y \perp M$, we want to arrive at a contradiction.
Then $\langle y, x \rangle = 0$, but also we know that $\langle 0, x \rangle = 0$ so $\langle y, x \rangle = \langle 0, x \rangle$ hence $y = 0$ but y was different from 0, a contradiction.
Therefore M is total in H. $\square$